



The Most Cost-Effective decoding platform SoC

Overview

F133 family is an advanced application processor designed for the video decoding platform. It integrates a 64-bit processor with RISC CPU instruction architecture to provide the most efficient computing power. F133 family supports full format decoding such as H.265, H.264, MPEG-1/2/4, JPEG, VC1, and so on. And the independent hardware encoder can encode in JPEG or MJPEG. Integrated multi ADCs/DACs and I2S/PCM/DMIC/OWA audio interfaces can provide the perfect voice interaction solution. F133 family supports rich display output interfaces to meet the requirements of the screen display in differentiated markets. F133 family can be used in network video machines, advertising machines, digital photo frames, car MP5, and so on.

The following is the F133 family differentiated features.

Features	F133-A	F133-B
CVBS IN	Not Support	Support

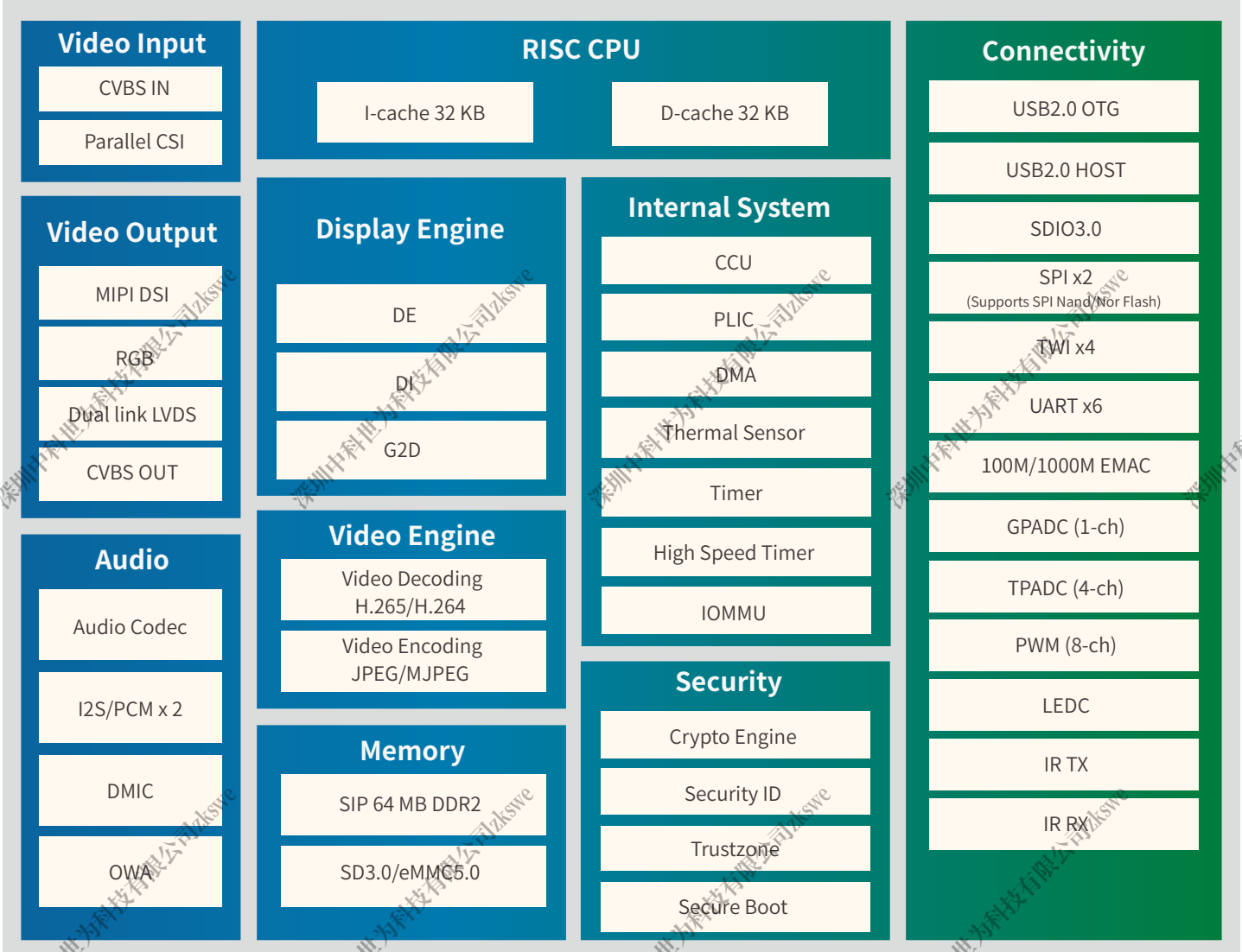
Highlights

- Integrated 64-bit RISC CPU processor provides powerful computing performance.
- The 1080p full format decoding, rich display output interfaces, and Allwinner SmartColor 2.0 display enhancement technology provide excellent video experience for users.
- To reduce the BOM cost, a 64 MB DDR2 die is embedded for F133 family.
- Rich peripheral interfaces, such as USB, SDIO, EMAC, TWI, UART, SPI, PWM, GPADC, IR TX&RX, and so on, greatly facilitate product expansion.
- The advanced process design with lower voltage and lower leakage, the power optimization design for typical scenes, and the enhanced heat dissipation package, improve the heating experience of the product.

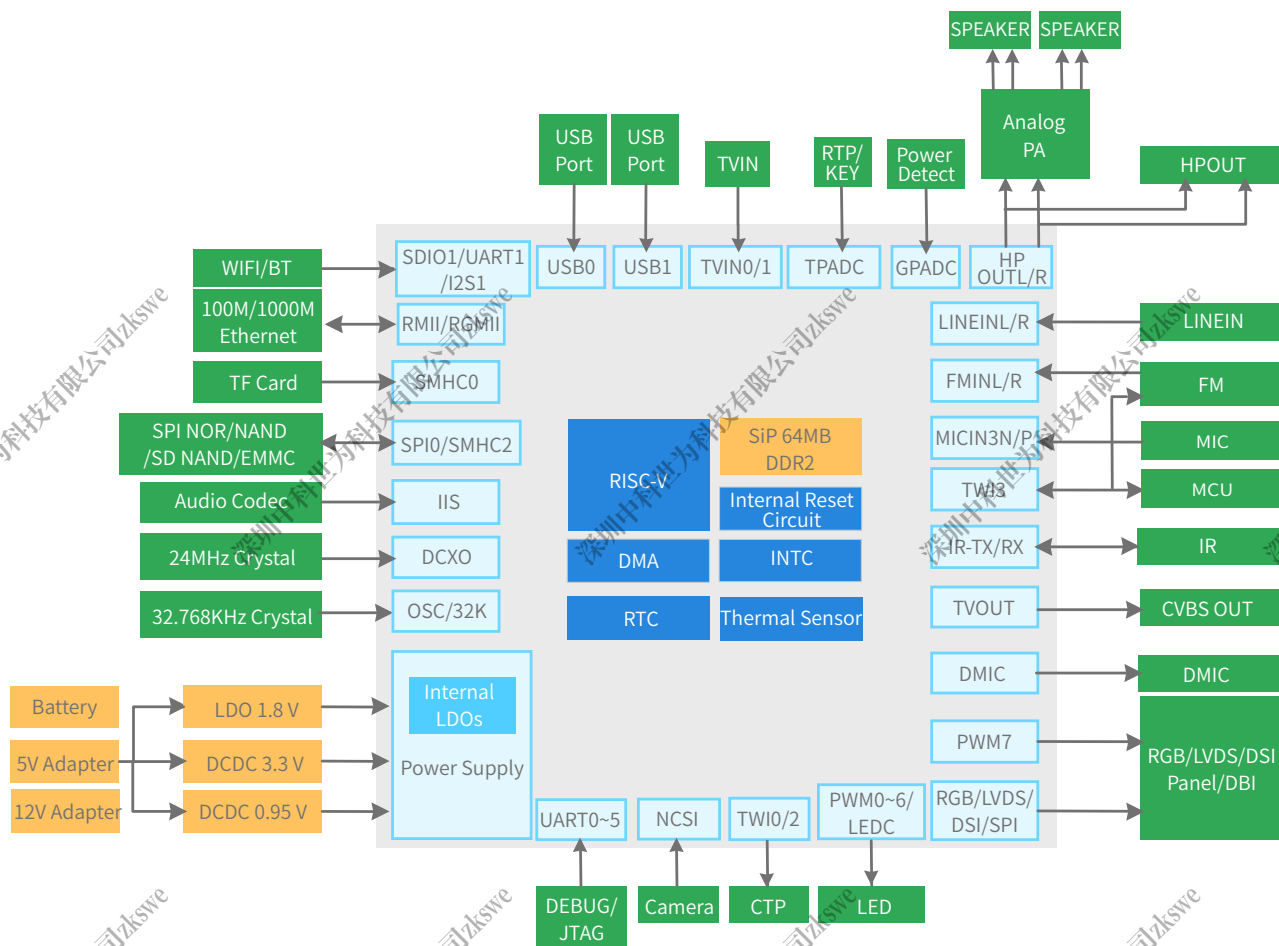
Features

CPU	• RISC CPU • 32 KB I-cache + 32 KB D-cache
Memory	• SIP 64 MB DDR2 • SD3.0/eMMC 5.0, SPI Nor/NAND Flash
Video Engine	• Video decoding -H.265 up to 1080p@60fps -H.264 up to 1080p@60fps -MPEG-1/2/4, JPEG, VC1 up to 1080p@60fps • Video encoding -JPEG/MJPEG up to 1080p@60fps -Supports input picture scaler up/down
Display Engine	• Allwinner SmartColor2.0 post processing for an excellent display experience • Supports de-interlace (DI) up to 1080p@60fps • Supports G2D hardware accelerator including rotate, mixer, lbc decompression functions
Video OUT	• CVBS OUT interface, supporting NTSC and PAL format • RGB LCD output interface up to 1920 x 1080@60fps • Dual link LVDS interface up to 1920 x 1080@60fps • 4-lane MIPI DSI interface up to 1920 x 1080@60fps
Video IN	• 8-bit parallel CSI interface • CVBS IN interface, supporting NTSC and PAL format
Audio	• 2 DACs and 3 ADCs • Analog audio interfaces: MICIN3P/N, LINEINL/R, FMINL/R, HPOUTL/R • Digital audio interfaces: I2S/PCM, DMIC, OWA
Security	• AES, DES, 3DES encryption and decryption algorithms • RSA signature verification algorithm • MD5/SHA and HMAC tamper proofing • Hardware random number generator • Integrated 2 Kbits OTP storage space
Connectivity	• USB2.0 OTG, USB2.0 Host • SDIO 3.0, SPI x 2, UART x 6, TWI x 4 • PWM (8-ch), GPADC (1-ch), TPADC (4-ch), IR TX&RX • 10/100/1000M EMAC with RMII and RGMII interfaces
Package	• eQFP128, 14 mm x 14 mm

Block Diagram



Application Diagram



ABOUT ALLWINNER

Allwinner Technology is a leading fabless design company dedicated to smart application processor SoCs and smart analog ICs. Its product line includes multi-core application processors for smart devices and smart power management ICs used by brands worldwide.

With its focus on cutting edge UHD video processing, high performance multi-core CPU/GPU integration, and ultra-low power consumption, Allwinner Technology is a mainstream solution provider for the global tablet, internet TV, smart home device, automotive in-dash device, smart power management, and mobile connected device markets. Allwinner Technology is headquartered in Zhuhai, China.

CONTACT US

For more product info, please contact service@allwinnertech.com, or scan the QR code to follow us on Wechat.

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